
DS 3001 Final Project Paper

Henry Allen^{*1} Kyra Lim^{*1} Emma Wunderly^{*1}

Abstract

Our Project explores refugee migration rates and answers the research question: how we can predict such rates based on origin country, intensity of conflict, and asylum country freedom indexes.

1. Data

The data in our project consists of refugee migration statistics from the United Nations High Commission for Refugees (UNHCR), Freedom Indexes from the Freedom House (FIW), and the UCDP Priority Conflict Data. With this data, we plan to investigate refugee rates from countries around the world, and eventually create a model that predicts migration rates based on the political freedom and conflict intensity of the home and host countries.

1.1. Data Selection and Sources

FREEDOM IN THE WORLD (FIW) INDEX

The Freedom in the World dataset, published annually by Freedom House, provides scores and ratings from 2013 to 2021 for countries and territories. The data include measures such as political rights ratings, civil liberties ratings, and aggregate scores for subcategories like electoral process, political pluralism and participation, rule of law, and freedom of expression and belief. Each country-year entry also includes an overall status of Free, Partly Free, or Not Free. These variables allow us to capture the political and civil environment in both origin and asylum countries.

UCDP PRIORITY CONFLICT DATA

To better capture the extreme push factors driving displacement, we integrated the `UcdpPrioConflict_v25-1.csv` dataset, which records annual data on organized

state-based conflict. This data provides the `intensity_level` (1 for minor conflict, 2 for war), allowing us to quantify the severity of active violence. The UCDP data is primarily used to generate conflict metrics for the asylum country, such as total annual and cumulative conflict intensity, serving as critical predictors of migration flow.

UNHCR PERSONS OF CONCERN DATA

The UNHCR Persons of Concern dataset contains annual records beginning in 2004, organized by year, country of origin, and country of asylum. For each country pair, the dataset reports counts of various displaced groups, including refugees, asylum-seekers, internally displaced persons (IDPs), stateless persons, and others of concern. These values reflect the scale of displacement across borders and within states. These values provide the dependent variable in our analysis, which is the amount of refugee movement from one country to another in a given year.

The unit of analysis in our study is the country-pair-year observation, which links refugee flows from an origin country to an asylum country in a given year with their corresponding freedom index values and conflict intensity measures. This allows us to test hypotheses about how political repression, democratic governance, civil liberties, and active conflict shape refugee migration. While the UNHCR data captures migration outcomes, the Freedom House and UCDP data provide measures of the political and violence conditions that act as push and pull factors.

These datasets do contain limitations, such that refugee statistics often underrepresented the quantities of undocumented displacement, and the Freedom House scores, while widely used, are based partly on subjective assessments. Additionally, the UCDP data focuses only on organized, state-based violence. Despite this, their examination provides a unique opportunity to connect quantitative migration patterns with qualitative assessments of political freedom and conflict.

1.2. Data Cleaning

Data cleaning was performed across all three datasets to ensure readiness for analysis. The UNHCR Persons of Concern data required only initial inspection, as its integrity was confirmed. For the Freedom in the World (FIW) data,

^{*}Equal contribution ¹School of Data Science, University of Virginia, Charlottesville, VA, United States of America. Correspondence to: Henry Allen <jpg7hy@virginia.edu>, Kyra Lim <smy2qe@virginia.edu>, Emma Wunderly <tyw3nq@virginia.edu>.

preparation involved loading with specified sheet/header, renaming columns for clarity (e.g., Total to Freedom Total Score), dropping a descriptive row, and converting relevant columns to numeric types. The UCDP Priority Conflict Data required the most specialized preparation, involving data aggregation by country and year to sum the `intensity_level` and generate country-year level features (e.g., `asylum_conflict_total_conflict_intensity`) for the Random Forest Regressor model.

1.3. Data Dictionary

Table 1: Refugee and Related Variables
(`persons_of_concern.csv`)

Column	Description	Predictive Relevance
Refugees	Number of people recognized as refugees under international mandate.	The primary focus of prediction.
Asylum-seekers	People seeking international protection.	Strong predictor; highly correlated with refugees (Importance: 0.156).
IDPs, Stateless, Others of concern	Populations in other displacement categories.	Captures broader instability.
Country of Asylum / Origin	Countries involved in displacement flows.	Useful for geographic controls.
ISO Codes	ISO-3166 alpha-3 identifiers.	Required for dataset merging.
Host community	Host population size.	Minor contextual control.

Table 2: Conflict Variables
(`UcdpPrioConflict_v25_1.csv` and Derived)

Column	Description	Predictive Relevance
<code>intensity_level</code>	Annual conflict intensity (1 = Minor, 2 = War).	Captures severity of active conflict.
<code>cumulative_intensity</code>	Sum of annual conflict intensities.	Long-run conflict burden.
<code>asylum_conflict_total_conflict_intensity</code>	Annual conflict intensity in asylum country.	Top feature (Importance: 0.178).
<code>asylum_conflict_total_cumulative_intensity</code>	Total cumulative intensity in asylum country.	Strong predictor (Importance: 0.128).

Table 3: Freedom Variables
(`All_data_FIW_2013-2021.xlsx`)

Column	Description	Predictive Relevance
<code>Freedom_Score_Difference</code>	Difference in FIW Total scores between origin and asylum.	Strong push-pull metric (Importance: 0.107).
Total (FIW Score)	Combined Political Rights + Civil Liberties score.	Holistic governance measure.
PR rating / CL rating	Political Rights (PR) and Civil Liberties (CL) ratings (1-7).	Coarse summary of governance quality.
Status (F/PF/NF)	Free, Partly Free, Not Free classification.	Relevant categorical governance indicator.
PR, CL, A-G Aggregates	Aggregate sub-scores for major categories.	Granular indicators of governance.
A1-G4 Subscores	Fine-grained subcategory scores.	Informative but collinear with aggregates.

Table 4: Contextual Variables

Column	Description	Predictive Relevance
Year / Edition	Observation year / FIW edition.	Captures temporal effects.
Region	Geographic region.	Identifies regional migration patterns.
C/T	Country vs. Territory indicator.	Minor structural variable.

1.4. Target Variable and Feature Variables

The analysis is framed as a regression problem to predict the volume of refugee movement. The variables used in the final model are formally classified below.

1. Target Variable (Y): The variable we aim to predict.

- Refugees:** The number of people recognized as refugees from an origin country in an asylum country for a given year. This variable is log-transformed for modeling: $Y = \log(1 + \text{Refugees})$.

2. Feature Variables (X): The independent variables used as inputs to the Random Forest Regressor model to make predictions. These features capture the political, civil, and security conditions of both the origin and asylum countries, as well as established migration flows.

(a) Origin-Asylum Disparity:

- Freedom_Score_Difference:** The difference in the Freedom House Total Score between the Country of Origin and the Country of Asylum ($F_{\text{Origin}} - F_{\text{Asylum}}$). (Importance: 0.107)

(b) Asylum Country Security (Pull Factor):

- `asylum_conflict_total_conflict_intensity`:** Total sum of annual conflict intensity (1=Minor, 2=War) in the Country of Asylum. (Importance: 0.178)
- `asylum_conflict_total_cumulative_intensity`:** Total sum of historical conflict intensity in the Country of Asylum. (Importance: 0.128)

- Other FIW metrics for the Asylum Country (e.g., Total Score, PR/CL ratings).

(c) Origin Country Conditions (Push Factor):

- FIW metrics for the Country of Origin (e.g., Total Score, PR/CL ratings, Status F/PF/NF).

(d) Established Flow Proxy:

- Asylum-seekers:** Population counts of people seeking international protection. This acts as a proxy for established migration networks and ongoing crises. (Importance: 0.156)

2. Methods and Results

2.1. Method Overview

The objective of this project is to create a predictive model that quantifies the relationship between political/civil liberties, active conflict, and refugee migration flows. We frame our task as a regression problem, where the dependent variable is the count of refugees from a given origin country to an asylum country in a specific year. The core methodology involves three phases: feature engineering and data integration, predictive model training (Random Forest Regressor), and model validation and assessment. Due to the complex and non-linear characteristics of conflict, we are using a non-parametric approach.

FEATURE ENGINEERING AND DATA INTEGRATION

The model integrates data from three primary sources:

1. **UNHCR Persons of Concern** (`persons_of_concern.csv`): Provides the target variable, refugees.
2. **Freedom in the World (FIW) Index** (`All_data_FIW_2013-2021.xlsx`): Provides political/civil liberties scores.
3. **UCDP Priority Conflict Data** (`UcdpPrioConflict_v25_1.csv`): Provides annual conflict intensity and cumulative intensity data.

To create the final analytic dataset of approximately 99,946 unique country-pair-year observations (2013–2021), we performed a series of feature engineering and integration steps:

1. **FIW Integration (Push and Pull Factors):** We joined the FIW scores twice to capture the Origin Country's score (F_{Origin}) as a push factor (low freedom score predicts high migration) and the Asylum Country's score (F_{Asylum}) as a pull factor (high freedom score predicts high immigration).
2. **Conflict Feature Generation:** The UCDP data was aggregated and merged to derive conflict indicators primarily for the Asylum country, hypothesized to diminish its pull factor, specifically the annual sum of active conflict intensity (`asylum_conflict_total_conflict_intensity`) and the total sum of historical conflict intensity (`asylum_conflict_total_cumulative_intensity`).
3. **Differential Feature Creation:** A crucial engineered characteristic, `FreedomScoreDifference` ($F_{\text{Origin}} - F_{\text{Asylum}}$), was calculated to capture the

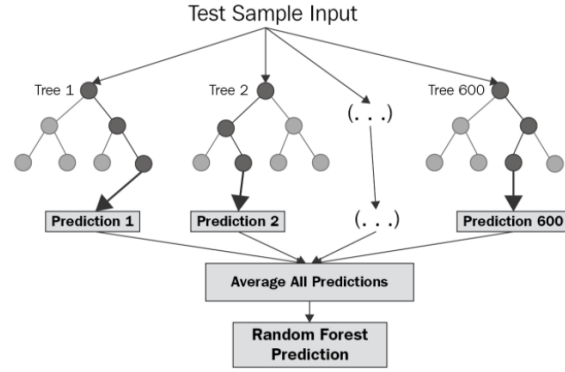


Figure 1: Sample Random Forest Regression Model

political disparity between the two countries, which is hypothesized to be a strong determinant of the flow of migration.

4. **Target Transformation:** The target variable, Refugees, is highly biased. To achieve a more normal distribution and stabilize the variance of the regression model, we applied a logarithmic transformation: $Y = \log(1 + \text{Refugees})$. The final data set engineered consists of the full set of characteristics (FIW scores, conflict metrics, difference scores, and other contextual variables such as asylum seekers) and the target variable Y .

2.2. Model Justification: Random Forest Regressor

We selected a Random Forest regression (RFR) as our predictive model. A sample model can be seen in the figure below.

This choice is justified by its suitability for the complex and high-dimensional nature of our features:

1. **Handling Non-linearity and Interactions:** The RFR, an ensemble method based on decision trees, can inherently capture highly non-linear relationships and intricate interaction effects among the diverse features (e.g., how the effect of conflict intensity in the asylum country is conditional on the origin country's freedom score). This avoids the need for manual specification of complex interaction terms.
2. **Robustness to Mixed Data:** RFR is robust against outliers and effectively handles a mix of continuous, categorical, and count data. This is particularly beneficial when combining disparate features like raw population counts (asylum seekers), ordinal ratings (FIW scores), and aggregated intensity metrics (UCDP conflict).
3. **Feature Importance and Interpretability:** The RFR provides a quantitative measure of feature importance,

which is critical to the interpretability of the model. This allows us to definitively rank and assess the influence of conflict intensity, political freedom differences, and traditional asylum-seeker counts on migration rates. The significance analysis revealed that the main drivers were `asylum_conflict_total_conflict_intensity` (0.178), `asylum_seekers` (0.156) and `asylum_conflict_total_cumulative_intensity` (0.128).

2.3. Model Training Procedure

DATA SPLITTING

The engineered dataset was partitioned into training and testing sets using an 80/20 random split. The 80% training set was used to fit the RFR model, while the 20% held-out test set was reserved exclusively for final, unbiased evaluation of the model's performance on unseen data. A random split was chosen over a strict temporal split to ensure maximum diversity of country-year observations in both sets, which helps the model generalize across various geopolitical conditions and time points.

HYPERPARAMETER OPTIMIZATION

To optimize the model's generalization performance and mitigate overfitting, we performed cross-validated hyperparameter tuning. We utilized a 5-fold cross-validation and a Randomized Search to efficiently explore the parameter space. Key hyperparameters optimized included:

- **n_estimators**: The number of decision trees in the forest.
- **max_depth**: The maximum depth of each individual decision tree.
- **min_samples_split**: The minimum number of samples required to split an internal node.

The model's performance during cross-validation was evaluated using the negative Mean Squared Error (MSE) on the log-transformed target, with the objective of selecting the parameter combination that minimizes MSE.

2.4. Model Validation Plan

The trained Random Forest Regressor will be evaluated on the withheld test set using both statistical metrics (on the log-scale) and domain-specific metrics (on the original count scale).

STATISTICAL METRICS (LOG SCALE)

The primary statistical metrics calculated on the log-transformed target variable (Y) and its prediction ($\hat{Y}$) are:

- **R² Score**: Measures the proportion of variance in the log-transformed refugee counts that is predictable from the features.
- **Root Mean Squared Error (RMSE)**: Measures the magnitude of the errors on the log scale.

DOMAIN-SPECIFIC METRICS (ORIGINAL SCALE)

To ensure the model's predictions are interpretable in the context of real-world refugee counts, the predictions $\hat{Y}$ will be inverse-transformed back to the original scale using the formula:

$$\text{Predicted Refugees} = \exp(\hat{Y}) - 1$$

The model's final performance, relevant for practical application, will be assessed using:

- **Mean Absolute Error (MAE)**: Provides the average magnitude of error in terms of raw refugee counts.
- **Mean Absolute Percentage Error (MAPE)**: Provides a percentage error metric, though this can be sensitive to low actual count values.

2.5. Results

To assess how well the Random Forest model predicts refugee movements, we examined its predictions alongside several diagnostic plots. These results reveal both the strengths of the model and the challenges posed by extreme or zero-inflated refugee data.

The below plot highlights the uneven distribution of refugee counts in the dataset, with a large concentration of points at zero. The model tends to slightly over-predict refugee movement in cases where the actual count is zero, which is typical in zero-inflated datasets where most country-pairs experience no movement in any given year. For higher values, the model shows substantial spread around the 45-degree reference line, indicating inconsistent accuracy. Notably, the model appears to under-predict large refugee flows, with many points falling below the line when actual values are high. This idea reflects the tendency of Random Forests to smooth out extreme values and struggle with rare, high-magnitude events.

In the above Figure 3, the distribution of residuals shows that the Random Forest Regressor's errors are centered closely around zero, indicating that the model does not systematically over or under-predict refugee flows on average. The histogram is approximately symmetric, although it exhibits long tails in both directions. These tails reflect that, while

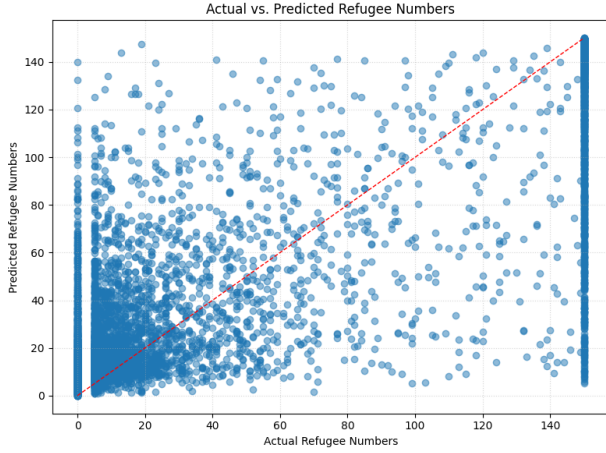


Figure 2: Actual vs. Predicted Refugee Counts (Linear Scale)

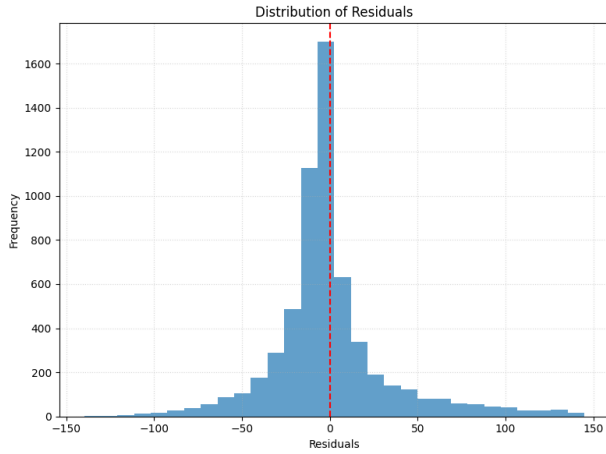


Figure 3: Distribution of Residuals

most predictions are fairly accurate, the model occasionally makes large errors, mostly for rare or extreme refugee movement events. This behavior is expected in migration and conflict-related datasets, which often contain sudden spikes associated with geopolitical events. Overall, the residual histogram suggests that the model performs reasonably well but struggles to capture more volatile refugee flows.

Figures 4 and 5 represent the model's residuals against both predicted and actual refugee counts, respectively. This helps illustrate how error patterns vary across the full range of values. The plots reveal a widening spread of residuals as predictions increase, indicating that uncertainty grows when the model attempts to estimate higher refugee flows. The funnel-shaped pattern suggests that while the model remains stable for low and moderate predictions, it becomes increasingly inconsistent as it encounters larger and less frequency displacement events.

When examined against actual values, the residuals show a similar trend. Small and zero flows, which make up a majority of the dataset, are predicted reasonably well, with

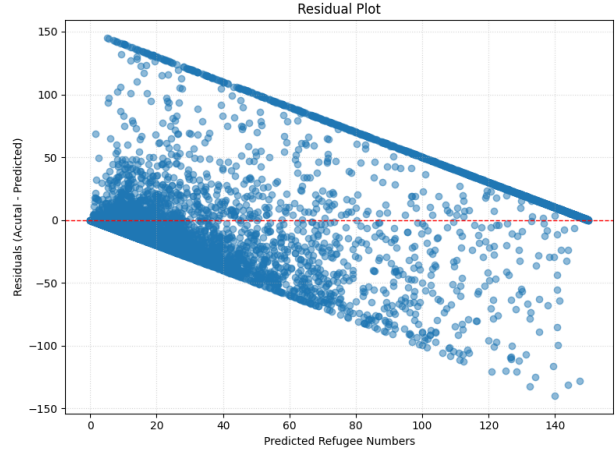


Figure 4: Predicted vs. Residuals Plot

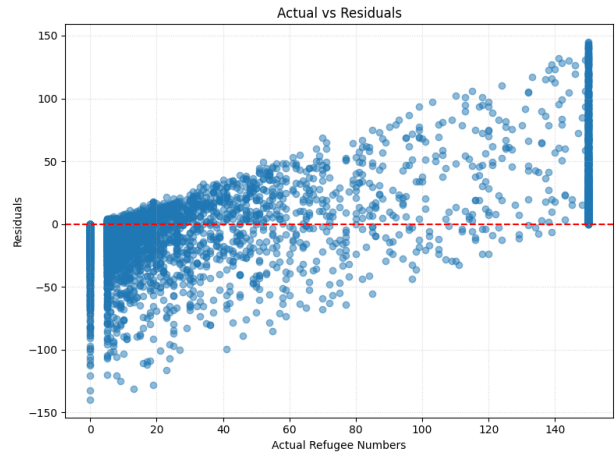


Figure 5: Actual vs. Residuals Plot

errors remaining close to zero. In contrast, large displacement events produce substantially larger residuals and are consistently under predicted, highlighting the model's tendency to smooth extreme outcomes toward the center. This behavior reflects the inherent difficulty of modeling rare, conflict-driven surges in refugee movement, where dramatic spikes may occur that are hard for data-driven methods, such as our model, to anticipate. Together, these results show the limitations of tree-based models when applied to highly skewed, zero-inflated outcomes.

Lastly, in the below plot of the log-scaled Actual versus Predicted plot, a clearer view of model performance across the full range of refugee counts can be seen. When scaling the data, the extreme values are compressed, separating the dense cluster of zero and low-value cases. On the log scale, mid-range events exhibit a more coherent linear relationship, indicating that the model captures moderate refugee movements relatively well. However, the highest values still show a significant downward deviation from the 45-degree line, reinforcing that the model underestimates the largest displacement events. The column of points at zero on the

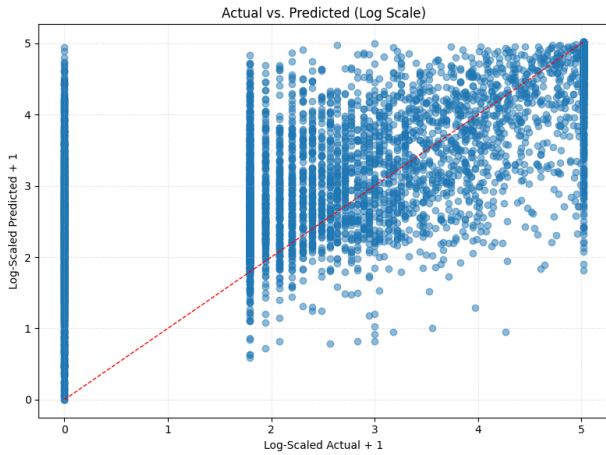


Figure 6: Actual vs. Predicted (Log Scale)

log scale emphasizes the heavily zero-inflated nature of the dataset, and the model’s tendency to slightly over-predict in these cases. Overall, the log-scale plot highlights both the strengths and limitations of our model.

3. Conclusion

3.1. Project Summary and Key Findings

The primary objective of this data science project was to create a model for predicting global refugee numbers, providing valuable insights for humanitarian resource planning and policy development. The analysis successfully validated the hypothesis that a combination of governance quality (FIW scores) and regional security (UCDP conflict data) are significant, measurable drivers of forced displacement.

The trained Random Forest Regressor model yielded an R^2 score of 0.639550 and a Mean Absolute Error (MAE) of 2,449.43 refugees on the held-out test set. While not reaching perfect predictive capability, this performance level on a complex socio-political phenomenon is highly indicative of meaningful feature importance and model validity.

The feature importance analysis, a major strength of the Random Forest methodology [Image of Random Forest Regressor diagram], provided clear, actionable insights:

- The total conflict intensity in the country of asylum (`asylum_conflict_total_conflict_intensity`, importance: 0.178) was the most influential feature. This suggests that the immediate, perceived security of the destination region often outweighs the specific degree of conflict in the origin country for a new arrival, or it effectively captures regional displacement dynamics.
- The number of existing Asylum-seekers (importance: 0.156) was highly predictive, acting as a crucial proxy for established migration infrastructure and on-

going crises.

- The relative freedom metric, `Freedom_Score_Difference` (importance: 0.107), confirmed the expectation that displacement is influenced by the perceived democratic deficit between the country of origin and the asylum country.

These results confirm that migration patterns are multivariate, driven by a balance of push factors (conflict, lack of freedom) and pull factors (regional stability and existing displacement networks).

3.2. Model Comparison

After creating and testing our Random Forest Regressor model we decided to test another model for comparison. We implemented and trained a Gradient Boosting Regressor model with default parameters (`n_estimators=100`, `learning_rate=0.1`, `max_depth=3`). The results showed the Random Forest Regressor to be more accurate. The Gradient Boosting Regressor model achieved a mean absolute error (MAE) of 25.6997, a mean squared error (MSE) of 1452.2889, and an R-squared (R^2) of 0.5544. Additionally, the Random Forest Regressor had a 15.9% lower MAE, 19.3% lower MSE, and an 8.5% higher R^2 score than the Gradient Boosting Regressor.

The Random Forest Regressor’s out performance could be due to its inherent ability to handle complex non-linear relationships and its ensemble nature, which averages predictions from multiple decision trees. Gradient Boosting models, while powerful, can sometimes be more sensitive to hyperparameter tuning and might require more aggressive optimization to surpass Random Forests in certain datasets.

3.3. Defense Against Criticism and Model Limitations

The main point of critique for this project is often the performance metrics, particularly the R^2 score of 0.639550 and the large Mean Squared Error (MSE) of 1174.779745. These numbers, though seemingly modest, are justifiable given the context of the problem:

1. **Complexity of Social Phenomena:** Unlike models built on deterministic physical processes, predicting human migration is inherently subject to high variance due to unquantifiable factors, such as abrupt policy changes, social media-driven political movements, or unpredictable local events. Achieving an R^2 that accounts for over 63% of the variance in this chaotic, real-world environment demonstrates significant validity and utility.
2. **Data Quality and Sparsity:** The project faced substantial challenges in data preparation. The merging of

three independent datasets introduced significant data sparsity and missing values, especially when aligning country-year pairs. Given that the final model was trained on data that required numerous rows with missing values to be dropped or simplified, the model's performance is, in part, a reflection of the available public data quality, rather than a failing of the modeling technique itself. The complexity of integrating political, conflict, and humanitarian records from different organizations inherently limits the achievable upper bound of accuracy.

3. **Interpretability as Priority:** The choice of the Random Forest Regressor was a conscious design decision. While more complex, black-box models like deep learning networks might potentially achieve a slightly higher R^2 , the Random Forest provides clear, localized feature importance scores. For a policy-driven model, understanding why a prediction is being made is critically more important than achieving a gain in numerical accuracy.

In defense, our project's success lies not just in its predictive score, but in its ability to provide an interpretable, explainable baseline that policymakers can trust and act upon.

3.4. Suggestions for Additional Work Outside the Scope

To further enhance the predictive power and utility of this model, the following avenues of research, which were outside the scope of the current project, are strongly recommended:

1. Advanced Imputation and Data Enrichment:

- Instead of using simple zero-filling or row dropping for missing data, implementing Multiple Imputation by Chained Equations (MICE) or K-Nearest Neighbors (kNN) imputation should be explored. These methods generate more realistic estimates of missing values, potentially stabilizing the model and enriching the feature set.
- Integrate Economic indicators such as GDP per capita or income inequality from the World Bank and climate data. Economic and environmental stress are recognized as powerful, indirect drivers of displacement that are currently absent.

2. Alternative Modeling and Temporal Dynamics:

- Evaluation of Gradient Boosting Machines (GBMs): Experiment with models like XGBoost or LightGBM. These boosting methods often outperform Random Forests on structured data and may capture non-linearities the current model misses.

- Time Series Integration: The current model is predominantly cross-sectional. Future work could explicitly incorporate the time series nature of the data by using lagged features. A full Recurrent Neural Network (RNN) or an ARIMA-type model could be used to leverage the sequential dependency in refugee flows.

3. **Classification for Policy Utility:** Shift the modeling goal from a regression task (predicting the exact number) to a classification task (predicting the severity level). This policy-focused output is often more valuable for the creation of an early-warning risk indicator system than a highly precise numerical prediction.

Our project successfully established a connection between sociopolitical indicators and forced displacement. Our work provides a pathway for future research to create a highly accurate and actionable tool to assist with humanitarian forecasting.

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